the top left panel, we show the individual data points in a scatter plot. In the remaining panels, we model the data using various numbers of variables, where each variable corresponds to a region inside the map of California. As the number of variables increases, the more fine-grained our model is. When this number is infinite, that is, when we can make a prediction in any region on this map, however small, our model exists in an infinite-dimensional space.

This is exactly what a Gaussian process is: a Gaussian distribution in an infinite-dimensional space. Strictly speaking, the concept of a Gaussian distribution doesn't apply when there are infinitely many variables, so the technical definition is the following.

IMPORTANT **Definition of a Gaussian process**

A Gaussian process is a collection of random variables such that the joint distribution of every finite subset of those variables is a multivariate Gaussian.

This means if we have a GP model to describe a function f, then the function values at *any* set of points are modeled by a multivariate Gaussian distribution. For example, the vector of variables [$f(1)\ f(2)\ f(3)$] follows a three-dimensional Gaussian; [$f(1)\ f(0)\ f(10)\ f(5)\ f(3)^{\mathrm{T}}$] follows a different, give-dimensional Gaussian; [$f(0.1)\ f(0.2)\ \ldots\ f(9.9)\ f(10)^{\mathrm{T}}$] follows yet another Gaussian.

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